Supplementary material

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Description. Supporting analyses for the main text: interval recalibration, rows outside the standard analysis set, a locked device forecast, and two replications on allometric scaling laws outside fusion.

For: "Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling" (L. Salcedo).

This document holds the supporting analyses referred to in the main text. Sections S1 to S3 are fusion analyses whose headline numbers are stated in the main text and whose construction, tables and caveats are given here in full: repairing the interval collapse and finding the limit of the repair, replicating the ranking inversion on rows the standard analysis set excludes, and a locked forecast at three device operating points. Sections S4 and S5 are the two replications outside fusion. Sections S6 to S8 are supporting material for results the main text states in prose: the three-kernel Gaussian-process ladder behind the mean-function ablation, the full model, kernel and split specification, and the per-label scores with the eligibility-threshold sweep. Sections S9 and S10 are two results the main text summarises in a paragraph each: a power law with a bounded correction, and the mixed model the clustered-validation literature pairs with this split design.

Why those two belong in a fusion paper is argued where they are first introduced, in the Discussion of the main text: HDB5 cannot separate the extrapolation failure from the ranking inversion, because on its nine features the flexible model always wins the interpolation split, and the nested predictor ladder of Sec. S5 can. Neither replication is evidence the confinement result depends on.

Section and equation numbers prefixed S refer to this document; unprefixed section names refer to the main text.

Each section here is a result the main text states and relies on. What moved is the construction, the tables and the caveats, not the finding.

S1 Repairing the intervals, and the limit of the repair

The diagnosis in the calibrated-intervals section of the main text makes a testable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves are tested below.

The label-level scheme uses an inner leave-one-label-out loop, and nothing in it touches the target label. It is called label-level rather than machine-level deliberately: the calibration unit is the database device/wall label, so for a target of JET-ILW the calibration set still contains JET. The device-level variant that removes that overlap is reported at the

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Table S1: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out database label (LOLO) and across the ITER-size-matched cut. The device-level counterpart is Table S2.

	split conformal		by label		+ distance	
Model	LOLO	cut	LOLO	cut	LOLO	cut
random forest	35%	3%	**88%**	29%	88%	40%
hist grad. boosting	45%	0%	**90%**	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	**95%**	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

end of this section. For each target m: remove m; the remaining labels are the training set. Inside that training set, hold out each label k with at least 30 rows in turn, fit on the others, and record $|\log \hat{\tau} - \log \tau|$ on k together with each of its rows' Mahalanobis distance from that inner fit set. Pool those residuals into the calibration quantile. Then fit the final model on all the training labels and score m once. Every conformity score therefore comes from a label its own model had not seen, which is the unit the test rows differ by, and no row from m enters any fit or any quantile. The distance-scaled variant fits $\sigma(d) = \exp(c_0 + c_1 d)$ to those calibration residuals by least squares on $\log|r|$ and divides the scores by it.

Neither scheme is a finite-sample-valid group-conformal procedure, and they are not claimed as one. Two things stand in the way. The conformity score is still a row, not a label, and rows arrive in correlated clusters, so the exchangeability the split-conformal proof needs is no more available here than in the calibrated-intervals section of the main text. And in the distance-scaled arm the scale function $\sigma(d)$ is estimated from the same calibration residuals it then normalises, which the standard argument does not cover. What these are is empirical held-out-label residual calibration: the quantile is the finite-sample conformal rank applied to those scores, and the coverage reported for it is measured rather than guaranteed. Recovering a guarantee would need label-level conformity scores and a scale model fitted on separate rows or cross-fitted, and neither is done here.

Both arms behave as the diagnosis predicts. On a held-out label, empirical row-level coverage returns close to nominal: every model lands within two points of it, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. Neither recalibration procedure tested restores near-nominal coverage there, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest).

The same calibration over physical devices. The scheme above calibrates on database labels, so for a target of JET-ILW the calibration set still contains JET. Collapsing the wall variants and repeating the whole procedure over the 11 physical devices removes that overlap: training, calibration and scoring then all use the device as the unit.

Table S2 says the diagnosis holds and the repair is less complete than the label-level numbers suggest. Plain split conformal is worse on a genuinely unseen device than on an unseen label, 31% against 35% for the forest and 38% against 45% for the booster, which is what removing the within-device overlap should do. Calibrating by device then recovers most of the shortfall for the forest (84%, and 88% with distance scaling, against 88% either

Table S2: The same three schemes with wall variants collapsed, so every calibration unit is a physical device and the held-out device is genuinely unseen. Pooled row-level coverage of nominal 90% intervals, with the median multiplicative half-width in brackets.

Model	split conformal	by device	+ distance
random forest	31% (1.23×)	84% (2.46×)	88% (2.76×)
hist grad. boosting	38% (1.24×)	77% (2.29×)	80% (2.53×)
ridge, log-linear	85% (1.34×)	92% (1.40×)	93% (1.43×)
power law, collisionless	88% (1.34×)	92% (1.39×)	93% (1.42×)
IPB98(y,2)*	90% (1.39×)	90% (1.39×)	90% (1.39×)

*historical reference, not refitted within these folds.

way at label level) but noticeably less for the booster, which reaches 77% and 80% where the label-level scheme reported 90% and 89%. The label-level number therefore overstates how well this repair calibrates the booster on a device it has never seen. Both linear models are unaffected within a point or two. The cost is visible in the widths: the trees buy that coverage at roughly 2.3 to 2.8× multiplicative half-width against 1.23× under plain split conformal, where the linear models stay near 1.4×.

Across the size cut with the same device grouping the picture is unchanged in kind. The forest and booster remain far from nominal at 41% and 12% by device and 57% and 21% with distance scaling, better than the label-level 29% and 4% but nowhere near 90%, while the constrained power law holds at 95% and 93%. Changing the calibration unit cannot repair a shift the calibration set does not contain, which is the point of the section.

Among the models refitted within folds, the constrained power law of the constraint section of the main text is the only one in Table S1 whose plain split-conformal coverage stays near nominal at the ITER-size-matched cut, and its intervals hold there under every scheme. IPB98(y,2) also holds at 88%, but it is a historical reference rather than a fold-fitted competitor.

S2 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5's 6228, and it is pinned by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 labels, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, scored against ITER89-P [1] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one failure mode that would not announce itself. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, that arm becomes a superset of the data the ranking-inversion section of the main text was computed on, and it reproduces the ranking-inversion section of the main text by construction. The matcher is therefore checked positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% over that law reported above and computed on rows it never saw, and the best model on an unseen label is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen label is the collisionless power law of the constraint section of the main text. Counting label–model pairs where a tree ensemble loses to the published law on an unseen label: 19 of 24 and 10 of 10; against the fold-refitted power law instead, 16 of 24 and 9 of 10.

Row disjointness is not discharge disjointness, and the paper's own argument says the discharge is the unit that matters. Of the 5358 H-mode rows, 425 (7.9%) are slices of one of 336 discharges that STD5 samples at a different time. Dropping those discharges outright leaves 4933 rows across 2864 discharges that share no shot with STD5 at all, and the arm survives: the best cross-validated model still beats the published law, by 39% against 42% on the full arm, and the best model on an unseen label is still IPB98(y,2), with the tree ensembles at 0.526 and 0.552 against the power law's 0.332. The reversal does not depend on the shared discharges.

So the reversal is not an artefact of the standard set's selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

S3 Locked predictions at three device operating points without matching H-mode observations

Everything above is retrospective. To make the disagreement checkable rather than merely described, what each model predicts is recorded for three real devices, with intervals, a date, and a digest over the input rows so that a later edit is visible. The parameter sets are published design values, and for SPARC and ITER, evaluating IPB98(y,2) on the adopted design vector reproduces the reference confinement time its own source quotes, to 0.6% and 2.9% respectively; the JT-60SA source supplies operating parameters rather than a quoted confinement time. Those are checks that the right machine is being described. None of the three has a *measured* H-mode thermal confinement time at the operating point tabulated here, which is what makes these predictions rather than retrodictions.

The ITER row is taken from the 1999 basis [2], and a forecast whose whole purpose is to be checkable later should say why it is locked against a document of that age. The eight engineering values it supplies for the $Q = 10$ inductive scenario, $R = 6.2\,\mathrm{m}$, $a = 2.0\,\mathrm{m}$, $I_p = 15\,\mathrm{MA}$, $B_t = 5.3\,\mathrm{T}$, $\bar{n}_e = 10 \times 10^{19}\,\mathrm{m}^{-3}$, $P = 87\,\mathrm{MW}$, $\kappa = 1.7$ and $M = 2.5$, are the ones the 2007 revision of the confinement and transport basis carries forward for the same scenario [3], so the choice is not a 1999 number that a 2007 number would replace. ITER's programme has been rebaselined since, on a schedule and a staging this paper does not attempt to summarise; the 2024 baseline and the research plan built on it are described by Loarte *et al.* [4], and the row below should be read as conditional on the operating point tabulated above rather than on ITER's current plan of record. It is not a prediction of whatever ITER's first $Q = 10$ discharge turns out to be, and a reader who prefers a different operating point can rescore every model at one: the complete input vector is in `results/forecast.json`.

Table S3: Predicted energy confinement time, in seconds, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest, so that the operating points these were computed at can be recovered later. Operating points are published design values: SPARC from [5] (V2 primary reference discharge), ITER from [2] ($Q = 10$ inductive baseline), and JT-60SA from its Research Plan [6] (full-current inductive scenario). Intervals are distance-scaled held-out-label calibration intervals, not the plain split-conformal intervals of the main text's coverage table, and are pointwise intervals for a new observation; their construction is specified in the text below. The complete input vector for each device is recorded in `results/forecast.json` under the content digest quoted below.

	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2)*	0.765 [0.58, 1.00]	0.479 [0.35, 0.66]	**3.591 [2.66, 4.85]**
power law, collisionless	0.724 [0.55, 0.96]	0.428 [0.31, 0.59]	2.837 [2.09, 3.84]
ridge, log-linear	0.720 [0.53, 0.98]	0.444 [0.32, 0.62]	2.858 [2.07, 3.95]
random forest	0.136 [0.03, 0.53]	0.449 [0.23, 0.87]	**0.435 [0.19, 0.98]**
hist grad. boosting	0.305 [0.14, 0.69]	0.418 [0.24, 0.74]	0.444 [0.24, 0.84]

How these intervals are built. They are not the plain split-conformal intervals of the calibrated-intervals section of the main text, and deliberately so: the main text's coverage table measures those covering 3% of rows across the ITER-size-matched cut, so quoting them for a next-step device would publish an interval already known to fail in exactly this situation. Each row instead uses the distance-scaled label-level scheme of Sec. S1, which gives the best size-cut calibration for the models that fail most severely there, applied with the whole database as the training set. Two disclosures belong with that. The scheme was chosen after comparing the three in Table S1, so these intervals are exploratory rather than prospectively selected, in the same sense as the constraint rung of the constraint section of the main text. And it calibrates over database labels rather than physical devices, so these are not device-level conformal intervals: Table S2 measures what that distinction costs, and for the tree ensembles it is not negligible. The label-level scheme is retained here because it is the one the forecast was locked under on 31 August 2026, and re-deriving the intervals afterwards would make the lock meaningless. A reader who prefers the device-level calibration should read the tree rows of Table S3 as correspondingly optimistic in width. For a given model, each eligible label k is held out in turn, the model is refitted on the remaining labels, and the absolute log residuals on k are recorded together with each of its rows' Mahalanobis distance d from that inner fit set. A scale $\sigma(d) = \exp(c_0 + c_1 d)$ is fitted to those pairs by least squares on $\log|r|$, dropping exactly-zero residuals; the conformity scores are the residuals divided by $\sigma(d)$; and the half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest such score at $\alpha = 0.10$, the same finite-sample rank used in the calibrated-intervals section of the main text. The interval reported for a device is that quantile multiplied by $\sigma(d)$ evaluated at the device's own distance, applied symmetrically in $\log \tau$ and exponentiated. The model whose point prediction is tabulated is then fitted once on all 6228 rows. For IPB98(y,2) the same construction is used on its analytic residuals, so its interval is comparable to the others rather than merely coexisting with them. Every interval is therefore an empirical pointwise calibration interval for a new observation rather than a finite-sample-valid conformal interval. Three things stand in the way, and only the last is about ITER: the conformity scores remain clustered row-level scores, the distance scale is estimated from the calibration residuals it normalises, and every calibration unit is smaller than ITER along the size direction, which is the limitation

Sec. S1 measures directly.

Table S3 illustrates the practical consequence of the disagreement in one place. On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**.

For the random forest that gap cannot be closed by tuning the standard estimator, and the ranking-inversion section of the main text says why: it predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s, and no hyperparameter moves that ceiling. The gradient booster carries no corresponding arithmetic bound but produces a similarly low point prediction. Because its prediction cannot exceed the largest training target, 1.321 s, the random forest cannot reach the 2.84 s to 3.59 s range the linear and physics-informed models predict for this ITER operating point, and its nominal 90% interval there runs from 0.19 s to 0.98 s, containing none of those predictions.

The JT-60SA column becomes checkable first, though not against any measurement published as of the 31 August 2026 forecast lock: the first campaign ran L-mode divertor plasmas, and its published power balance reports the energy confinement time following L-mode scaling [7], whereas the row above is an H-mode prediction scored against IPB98(y,2). Checking it needs H-mode operation at a comparable design point, which the research plan [6] places in a later campaign and which the overview of first operation and forward plans [8] discusses in view of ITER and DEMO; no date is put on that here, because the schedule is not ours to assert and the prediction is locked against a parameter vector rather than against a calendar.

Each row of Table S3 carries its own interval and is assessed separately against it, so the range quoted here names one model rather than the set. A stationary JT-60SA H-mode near this design point reporting a thermal confinement time outside 0.349 s to 0.657 s would fall outside the IPB98(y,2) reference row's stated 90% predictive interval. The fitted rows are assessed at their own bounds, which are narrower for the constrained power law (0.313 s to 0.587 s) and wider for the random forest (0.232 s to 0.867 s). A single observation outside a nominal 90% interval is evidence against that forecast rather than a refutation of it: a perfectly calibrated model still places about one observation in ten outside such an interval, so a single miss is inconsistent with the prediction at the stated interval level and no more. Because all five intervals overlap across roughly this range, a measurement inside it separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label in the database tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

S4 The same audit on a scaling law from another science

Every result in the main text rests on ITPA data, which leaves open whether any of it is a fact about fusion specifically. Mammalian basal metabolic rate against body mass is the same object and the older one: Kleiber [9] found in 1932 that metabolic rate scales as mass to the 3/4. Whether the true exponent is 3/4 or 2/3 has been argued for decades

Table S4: Kleiber's law under the same three splits. Scores are log-RMSE.

Model	CV, by species	leave-one-order-out	widest mass cut
Kleiber, exponent 3/4	0.396	0.440	**0.374**
power law, free exponent	0.374	0.432	0.496
hist grad. boosting	0.418	0.487	0.889
random forest	0.437	0.517	0.710

and this paper takes no position; 3/4 is used here as the canonical historical reference law a fitted model has to beat. Taxonomic *order* is the unit a species belongs to, as a tokamak is the unit a discharge belongs to, and *body mass* is the axis a next instance lies beyond, as machine size is. The data is Supplement 1 of a metabolic-rate compilation deposited on Figshare (article 3549807), pinned by SHA-256 as the HDB5 files are, giving 541 species records across 11 orders whose median masses span 342x. The analysis runs through the domain-agnostic module released with this work rather than a copy of the fusion pipeline, so it is the procedure a reader would apply to their own data, and the problem has a *single predictor*: no model can win by finding a better combination of features, so the only thing separating them is functional form. Refitting the exponent freely gives 0.687 against Kleiber's 0.75, so the constraint is not free here either.

Table S4 reports the three splits and Figure S1 shows them.

The extrapolation failure reproduces. Both tree ensembles are worse than both power laws at all 8 mass cuts, and the random forest loses to Kleiber on 9 of 11 held-out orders. Two signatures reproduce with it. Tree error rises with extrapolation distance, at $\rho = +0.64$ and $+0.65$ against $+0.39$ and $+0.45$ for the power laws, which is the same sign as the distance diagnostic in the main text but a much narrower contrast, since here the power laws rise with distance too. And the training-range bound binds at every cut.

The ranking reversal does not, and that bounds the claim. The trees never win the easy split either: under cross-validation by species they score 0.437 and 0.418 against the free power law's 0.374. With no inflated margin there is nothing available to invert. The HDB5 gain came from a flexible model exploiting structure across nine features; with one predictor and a relationship close to a straight line in logs, a tree has far less to exploit and does not buy the interpolation win that later reverses. So the inversion tracks whether the flexible model wins interpolation first, which no single database could have shown. Whether predictor count is itself the operative variable is a separate question this comparison does not settle.

A limitation of this replication. The 11 taxonomic orders are not statistically independent: they share phylogeny, so two orders that diverged recently are more alike than two that did not, and the held-out group is therefore not a draw from a pool of independent units. Comparative biology normally handles this with phylogenetically controlled regression [10]; this audit does not. It treats orders only as held-out groups, and models no phylogenetic covariance. What that costs is any rate quoted here: the counts below establish a direction, not a frequency with which a constraint helps.

The constraint result reproduces in direction, not in strength. Kleiber's published exponent costs +0.023 under cross-validation and wins the widest cut, 0.374 against

0.496, where the held-out order is 69x heavier than the training median, which is exactly the trade the Connor–Taylor constraints make in the main text. But it is weaker, winning at 4 of 8 cuts rather than 8 of 8, decisively at the extremes and narrowly losing in the middle. That a constraint helps out of distribution replicates; that it helps reliably does not.

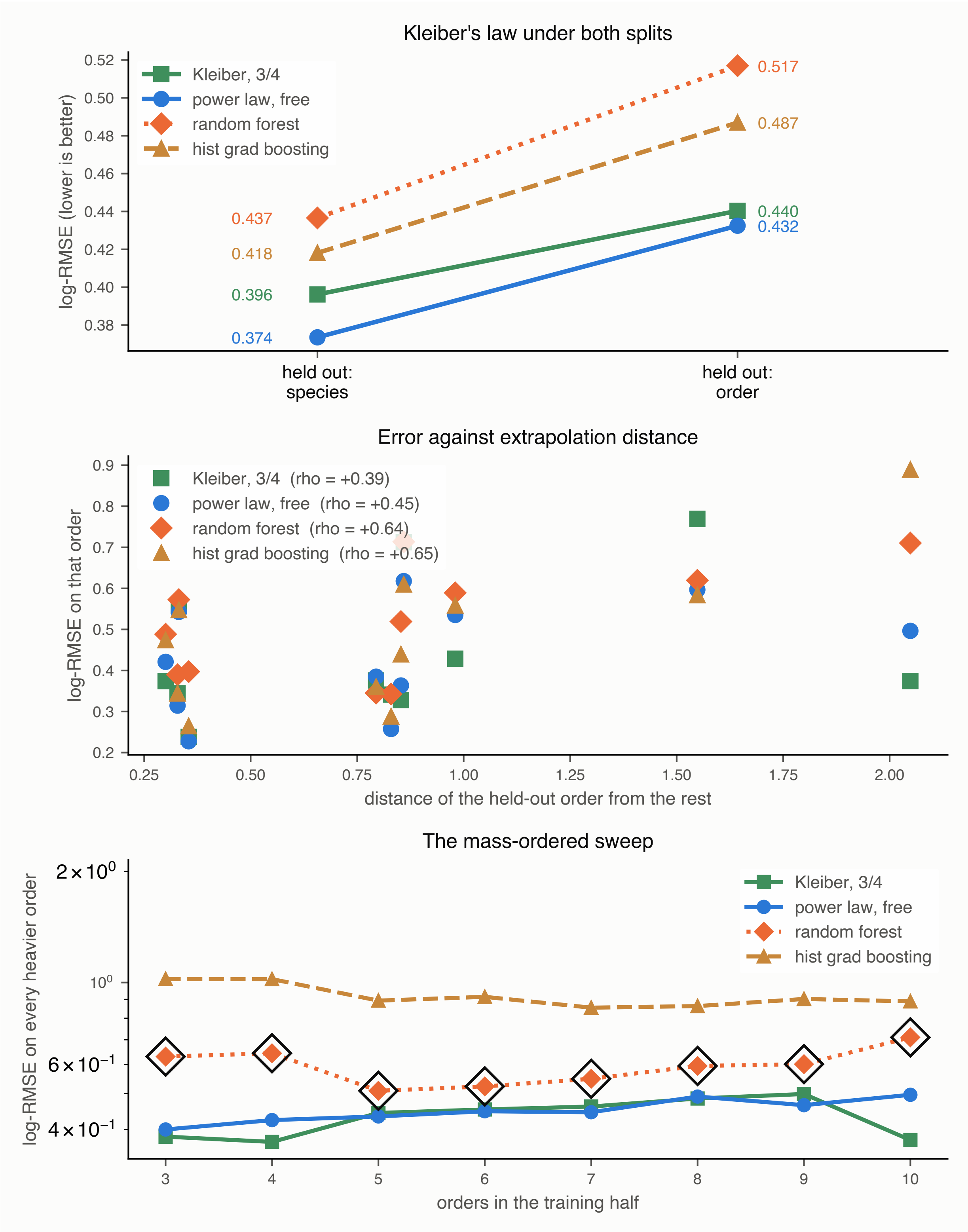

Figure S1: Kleiber's law under the same three splits. Top: no inversion, because the trees lose under both splits and so have no interpolation margin to give up. Middle: error against extrapolation distance, rising for the flexible models as it does on the ITPA H-mode database. Bottom: the mass-ordered sweep, with the circled points marking cuts where the random forest is held inside its training range.

S5 The reversal's precondition, measured directly

Sec. S4 leaves an asymmetry. The extrapolation failure reproduced in full on Kleiber's law; the ranking inversion did not, and the explanation offered there was a conjecture: with a single predictor a tree has little to exploit, never wins the interpolation split, and so has no advantage to lose. Two datasets cannot test that, because HDB5 and the mammalian compilation differ in feature count and also in science, group structure, sample size and noise.

A third dataset can probe that with a nested predictor ladder on the same rows and splits. The Biomass And Allometry Database [11] records, for individual plants, a nested set of measurements: basal diameter, height, leaf area and leaf mass. Taking the rows complete in all four gives 3599 plants across 53 species, 33 of them with the 30 individuals required to hold a species out, spanning 36x in diameter across species medians and over seven orders of magnitude in mass. It is released CC0 and pinned by SHA-256 as the other deposits are. Species plays the part of tokamak, diameter the part of machine size, and the branching-network prediction that mass scales as diameter to the 8/3 [12] the part of IPB98(y,2): an exponent derived from theory rather than fitted, which refits here to 2.512 and costs 0.5641 against 0.5780 in sample when held.

Each rung of the ladder is a prefix of the next and all four are scored on the same rows. The rows, the splits and the model specifications are held fixed; each rung adds one named predictor to the nested feature set. Predictor count is therefore what is varied, but not the only thing that changes with it, since each added column carries its own biological information. The split structure matters and is easy to get wrong: grouped cross-validation by discharge keeps every machine on both sides, so its analogue keeps every species on both sides. Grouping the cross-validation by species instead would be a second leave-one-group-out, and against it no reversal can appear at any dimension.

Table S5 and Figure S2 separate two statements that had been travelling as one. The interpolation margin rises monotonically and crosses zero between two and three predictors. The extrapolation margin does not shrink as the trees improve; it widens, and the power law wins the held-out species at 4 of 4 rungs. The inversion therefore appears at 3 predictors and every rung above, which is exactly where the interpolation margin turns positive. What the ladder shows is that the inversion tracks the interpolation margin, not that three is a threshold: each new column adds real biological information as well as a dimension, and leaf mass in particular is closely related to the target, so the rungs differ in more than their count.

So the extrapolation failure reproduces in all three datasets tested here, at every predictor count tested. In this ladder the extrapolation disadvantage persists at every rung, while the ranking inversion appears only once the flexible model gains an interpolation advantage. The inversion is therefore not necessary for the extrapolation failure, and is better read as a visible symptom of it than as the thing itself. At the one- and two-predictor rungs the same qualitative extrapolation hazard is present without an interpolation advantage whose loss would signal it.

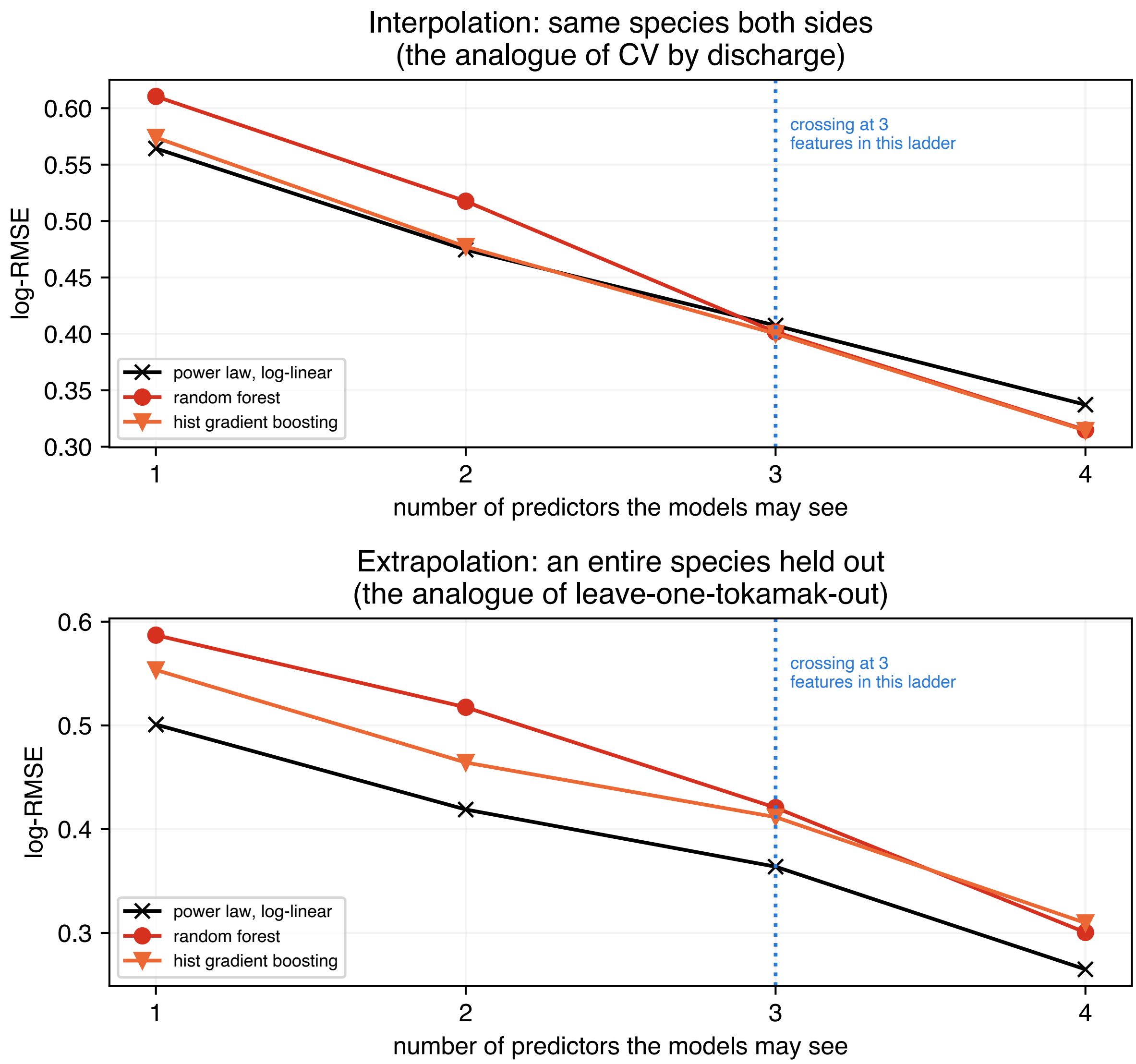

Figure S2: The reversal's precondition, on 3599 plants held fixed. Top: the interpolation split, where every species appears on both sides. Bottom: an entire species held out. The rows, splits and model specifications are held fixed; each rung adds one named predictor to the nested set. The top curves cross; the bottom ones do not.

Table S5: The best tree ensemble's margin over the log-linear power law, on 3599 plants held fixed across the four rungs. Positive favours the trees. The interpolation column crosses zero; the extrapolation column does not, and grows slightly against them.

predictors	interp. gain	held-out gain	reversal
diameter	−1.7%	−10.5%	no
+ height	−0.6%	−10.8%	no
+ leaf area	+1.8%	−13.1%	yes
+ leaf mass	+6.8%	−13.4%	yes

S6 The three-kernel Gaussian-process ladder

The main text states the ordering across three Gaussian-process kernels and builds its mean-function ablation on it. The ladder itself is here: the same log features, the same optimiser and the same rows, differing only in which kernel terms exist. Table S6 gives the scores, and Figure S3 shows them beside the spread of each rung's predictions, which is where a model that has reverted to a constant becomes visible as such.

Specification. The three kernels are a squared-exponential with a single isotropic length scale shared across the standardised log features, a dot-product kernel, and their sum, each with an additive `WhiteKernel` for observation noise. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log \tau$: this is what makes "reverts to the prior" and "returns the average training target" the same statement, and the parallel with a tree ensemble exact. Hyperparameters are fitted by maximising the log marginal likelihood on a seeded subsample of 1000 training rows; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold. No held-out row enters either stage. Every rung starts from the same initialisation for every term it has, so the three fits differ only in which terms exist. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term, exponentiated to τ. The kernel equations, initialisations and bounds are in Sec. S7.

The control behaves as intended: the linear-kernel process and ridge agree to 0.0000 under cross-validation and on a held-out label, and to 0.0007 at the ITER-size-matched cut, which is what licenses reading the other two rows as consequences of the kernel. The bounded rung then fails exactly as a tree does. It beats the power law under cross-validation and scores 1.948 at the ITER-size-matched cut, worse than predicting a constant, with error tracking distance at $\rho = +0.65$. The mechanism is the one in the main text: a random forest cannot exceed the largest training target, and an RBF process reverts to its prior mean. Both are structural long-range behaviours rather than tuning shortfalls, and on the held-out rows this one's predictions carry half the spread of the truth.

Adding a dot-product term to that same kernel keeps the nonlinear RBF component it already had and supplies an unbounded linear trend alongside it. Flexibility is retained rather than held exactly fixed, since a linear kernel enlarges the function class as well as changing the asymptote; what the comparison isolates is the availability of a continuing trend. It gives the best cross-validated score in this work, 0.112 against the random forest's 0.128; it scores 0.191 at the ITER-size-matched cut, better than the analytic law, which is a historical reference rather than a fold-fitted competitor, and second only to the constrained

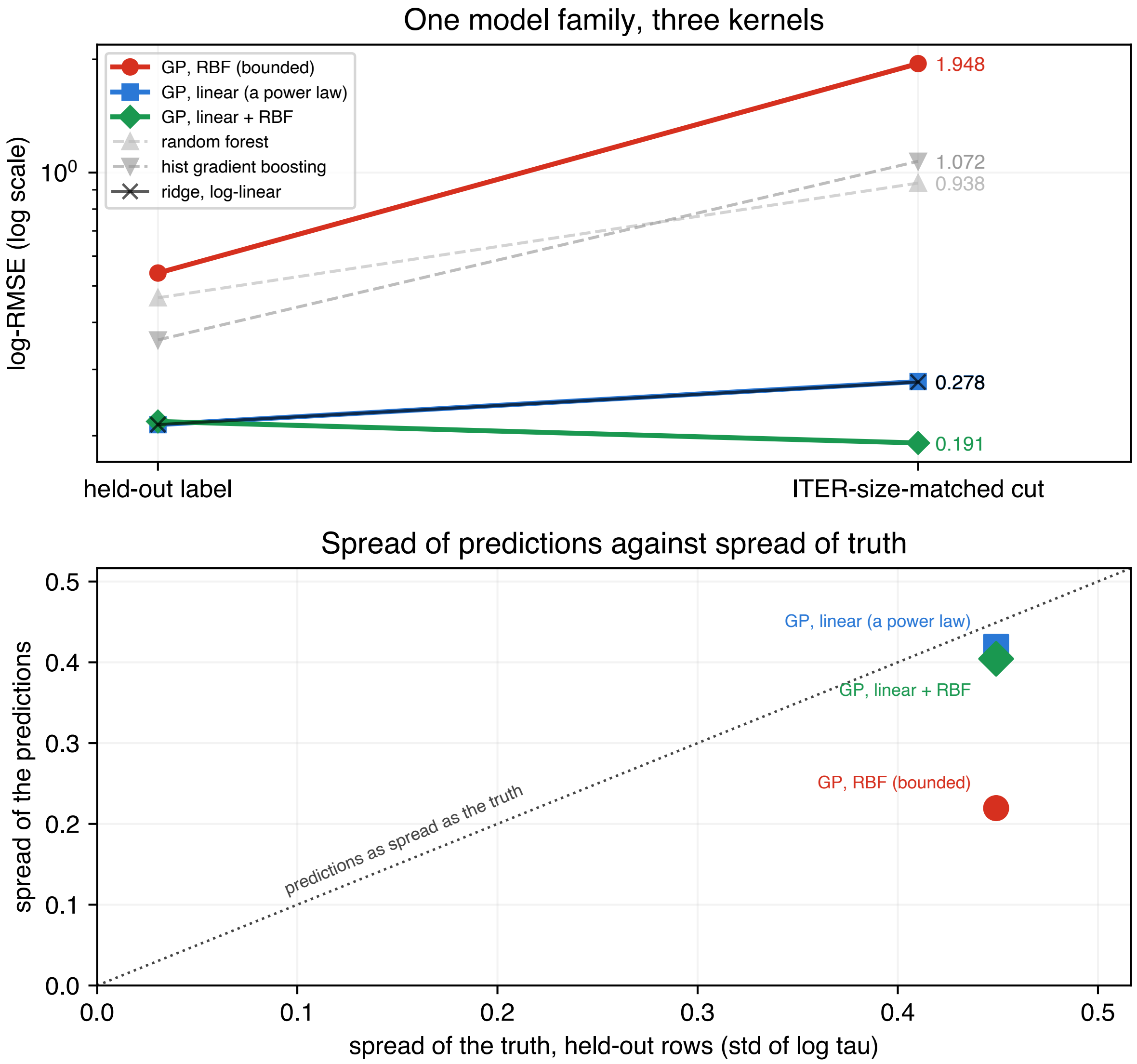

Figure S3: One model family, three kernels. Gaussian processes with a radial-basis-function (RBF) kernel, a linear (dot-product) kernel, and their sum. Top: log-RMSE on a held-out label and at the ITER-size-matched cut. The kernels differ in whether they permit a trend that continues at long range; adding the linear component also enlarges the function class. Bottom: spread of the predictions against spread of the truth on the held-out rows. A model whose posterior has reverted to its prior mean predicts nearly the same value everywhere, so its spread sits near zero however good its in-range fit was: the same failure a tree ensemble shows when it cannot exceed its largest training target, reached by a different route.

Table S6: log-RMSE under the three splits, and the rank correlation of per-label error against extrapolation distance. The intended difference across the first three rows is whether the kernel admits a trend that continues at long range; adding the linear term also enlarges the function class. The three span a factor of ten at the ITER-size-matched cut. IPB98(y,2) is a historical reference fitted to the DB2.8 predecessor of this database [13] and is not refitted within these folds.

model	CV	held-out	ITER cut	ρ
GP, linear + RBF	**0.112**	0.218	**0.191**	−0.01
GP, RBF only	0.142	0.541	1.948	+0.65
GP, linear only	0.181	0.214	0.278	−0.06
random forest	0.128	0.465	0.938	+0.85
IPB98(y,2), historical ref.	0.199	0.188	0.194	−0.49
mean baseline	0.869	0.994	1.459	+0.56

fit of the main text at 0.183; and its error is flat against distance at $\rho = -0.01$.

S7 Full model, kernel and split specification

This section carries the settings the main text points at. "Library defaults" is not a specification, and a default may differ in a later release, so the consequential settings are written out rather than referenced. Library versions are pinned in the repository's `constraints.txt` and named in the main text.

S7.1 The models of the headline comparison

The power law is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, chosen before any held-out score was computed and never varied. The random forest is `RandomForestRegressor` with 300 trees, `criterion="squared_error"`, unlimited depth, `min_samples_split` = 2, `min_samples_leaf` = 1, `max_features` = 1.0 and bootstrap resampling. That last setting means every split considers all nine features rather than a random subset, so this is bagged CART with bootstrap resampling and not the feature-subsampled forest of [14]; it is scikit-learn's regression default, it is a choice rather than an oversight, and the nested search of the main text moves off it in 13 of 19 folds. The gradient booster is `HistGradientBoostingRegressor` with squared-error loss, learning rate 0.1, at most 100 boosting iterations, 31 leaves, no depth limit, `min_samples_leaf` = 20, no L_2 penalty and 255 bins. Its `early_stopping="auto"` resolves to *off* at every sample size here, since that setting enables early stopping only above 10 000 rows and no training fold reaches that. The polynomial controls are `PolynomialFeatures` of degree 2 and 3 without a bias column, then the same scaler and ridge. Neither tree ensemble is standardised, which is immaterial to both.

S7.2 The Gaussian-process kernels

The three kernels are

$$\begin{aligned} k_{\mathrm{RBF}}(x, x') &= \sigma_r^2 \exp\Big(-\tfrac{1}{2}\|x - x'\|^2/\ell^2\Big) + \sigma_n^2 \delta_{xx'}, \\ k_{\mathrm{lin}}(x, x') &= \sigma_l^2 \, (x^\top x' + \sigma_0^2) + \sigma_n^2 \delta_{xx'}, \\ k_{\mathrm{lin+RBF}}(x, x') &= k_{\mathrm{lin}} + k_{\mathrm{RBF}} - \sigma_n^2 \delta_{xx'}, \end{aligned} \tag{1}$$

with a single isotropic length scale ℓ shared across the standardised log features, so the RBF component is isotropic rather than automatic-relevance-determining. Features are standardised inside the pipeline, so each fold's scaler is fitted on that fold's training rows alone and ℓ is in those units. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log \tau$: this is what makes "reverts to the prior" and "returns the average training target" the same statement, and the parallel with a tree ensemble exact. Observation noise enters as an additive `WhiteKernel` rather than through a fixed jitter. Hyperparameters are fitted by maximising the log marginal likelihood with scikit-learn's default L-BFGS-B optimiser, `n_restarts_optimizer` = 0, from the shared initialisation $\sigma_l^2 = 1$, $\sigma_0 = 1$, $\sigma_r^2 = 0.5$, $\ell = 1$, $\sigma_n^2 = 0.05$, under the bounds $\sigma_l^2, \sigma_0 \in [10^{-3}, 10^3]$, $\sigma_r^2 \in [10^{-3}, 10^3]$, $\ell \in [10^{-2}, 10^3]$ and $\sigma_n^2 \in [10^{-6}, 1]$. Every rung starts from the same values for every term it has, so the three fits differ only in which terms exist. The optimisation runs on a subsample of exactly 1000 training rows drawn without replacement by `numpy.random.default_rng(42)`, or on the whole fold when it holds fewer; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold, which is one Cholesky factorisation. No held-out row enters either stage. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term, exponentiated to τ.

S8 Per-label scores and the eligibility-threshold sweep

Both results are stated in prose in the main text. The tables are here for a reader who wants the per-unit numbers behind them: Table S7 gives the leave-one-label-out scores the distance correlation is computed over, Table S8 the sweep of the row threshold that decides which labels are scored at all, and Figure S4 the full size-ordered sweep whose matched cut the main text reports.

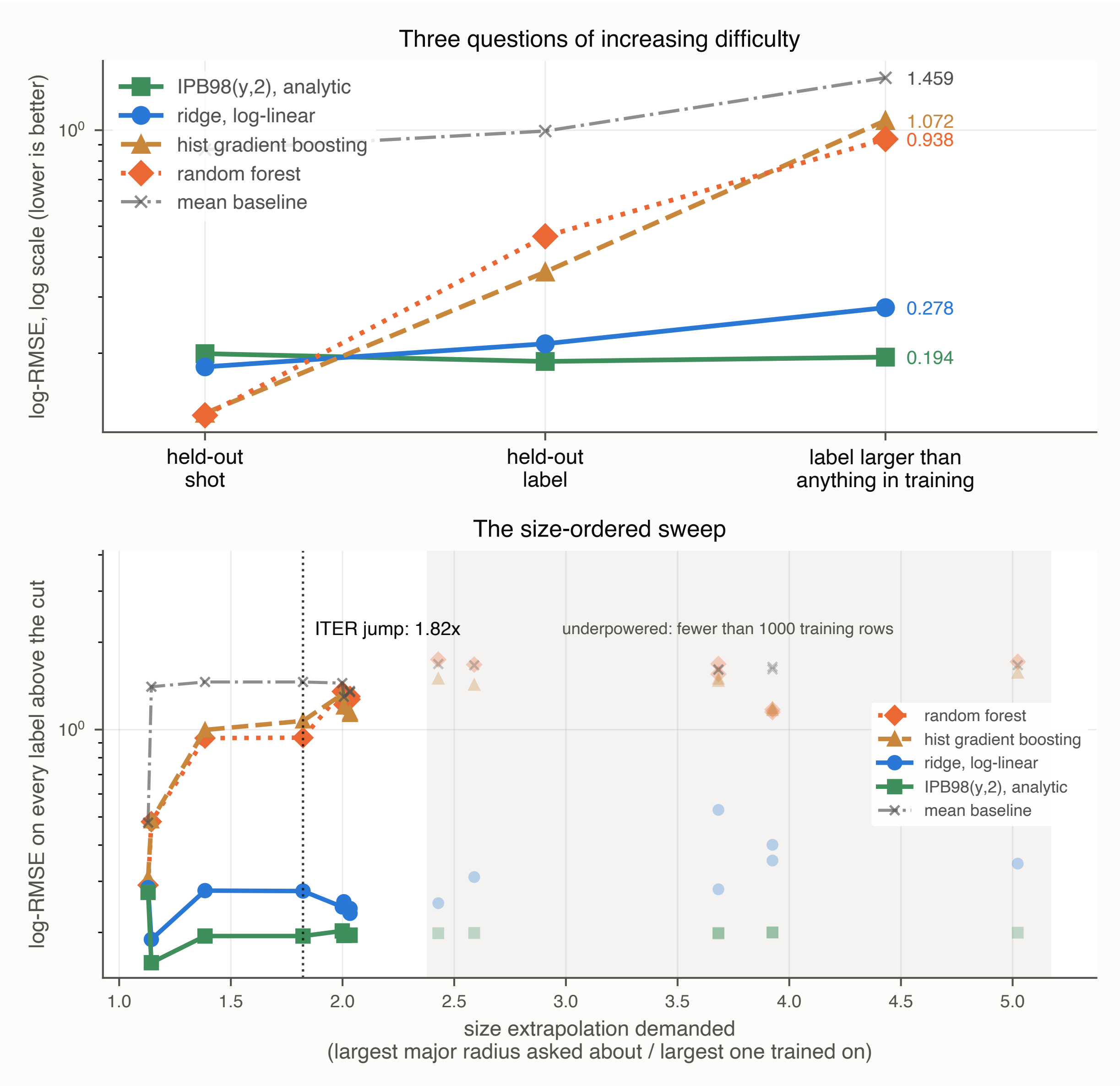

Figure S4: Size-ordered extrapolation. Top: the escalation across three splits of increasing difficulty, where the tree ensembles collapse toward the mean baseline. Bottom: the sweep across every cut, with underpowered cuts drawn faint and excluded from claims. The power law holds across the sweep and the trees do not recover.

Table S7: Per-label log-RMSE under leave-one-label-out, ordered by Mahalanobis distance of the held-out label's mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular: the design matrix is rank deficient by two, as the main text shows). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

Table S8: Leave-one-label-out at four eligibility thresholds. Only the set of labels scored changes across rows; the training folds and the models are identical throughout. The 30-row threshold is the one reported elsewhere in this paper. p is the exact two-sided sign test on the win count.

min. rows	labels scored	power law	forest	forest worse on	p
10	15	0.262	0.473	14 of 15	9.8×10^{-4}
20	13	0.214	0.465	13 of 13	2.4×10^{-4}
30	13	0.214	0.465	13 of 13	2.4×10^{-4}
50	11	0.223	0.449	11 of 11	9.8×10^{-4}

S9 A power law with a bounded correction

The diagnosis in Sec. 4.1 of the main text suggests a specific repair. Fit the power law, learn a correction on its log residuals, and damp it by λ:

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}} . \tag{2}$$

At $\lambda = 0$ this is exactly the ridge row of Table 1 of the main text. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [15], and this section makes no claim on that idea. What is varied is the one property the diagnosis points at, the long-range behaviour of the correction: one corrector is a degree-2 log polynomial, which is unbounded, and the other is depth-2 boosted trees, which never left the residual range they were trained on here. The two families differ in more than that, and the controlled version of the comparison is Sec. 12.1 of the main text. The damping factor is swept over nine values from 0 to 1, both correctors carry deliberately strong penalties so that λ rather than their own regularisation does the damping, neither corrector's hyperparameters are tuned, and the whole estimator is refitted from scratch inside every training fold; the exact specification is in `analysis_hybrid.py` in the archived code.

Across the ITER-size-matched cut the two correctors go opposite ways. Both start at

0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out labels and by the widest margin on the most distant of them, JT-60U (0.440 to 0.281).

The mechanism is measurable. Fitted on the 14 small labels, the base power law is *biased* on the held-out ones, at a mean log residual of -0.218, so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves its training range on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same bounded range that makes these ensembles useless as predictors of τ on a larger machine limits how much extrapolation error the correction can contribute, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero. Bounded is not the same as right: a correction confined to that range can still point the wrong way, and what is measured here is that on these rows it does not.

Two limits qualify every later summary of this section. Cross-validation selects $\lambda = 1$ in both families while the rung best on a held-out label is $\lambda = 0$, so a practitioner tuning on the split they would ordinarily use does not arrive at the rung that transfers. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, the bounded hybrid equals the power law at $\lambda = 0$, where it is that model, and beats it at all eight non-zero damping settings swept there, with the mechanism measurable.

S10 The estimator this split design is usually paired with

Every comparison above sets a pooled global fit against ensembles. The clustered-validation literature cited here for the split design [16] pairs that design with a third kind of estimator: a mixed model carrying a random intercept per device, from which a device never seen is predicted by the population-level fixed effects alone. It is neither a power law nor a tree, it is the fit matched to the deployment question, and it is scored here rather than conceded.

Write $\log \tau_{ij} = x'_{ij}\beta + u_i + \varepsilon_{ij}$ with $u_i \sim N(0, \sigma_u^2)$ over devices, fit the variance components by restricted maximum likelihood, and predict a held-out device as $x'\beta$. One quantity separates this from the pooled fit, the ratio $\lambda = \sigma_u^2/\sigma^2$. At $\lambda = 0$ the estimator *is* the pooled power law, and it reproduces the ridge column of Table 2 of the main text to three decimals by label, at 0.214, and to two by device, at 0.212 against 0.211: the closed-form least squares this estimator reduces to and the ridge spelling of the same fit part company in the fourth decimal, which is the size of the penalty and not of a disagreement. As λ grows the intercept absorbs more of the between-device variation, which leaves β identified increasingly by variation within a device; major radius and aspect ratio are constant within a device by construction, and elongation and isotopic mass nearly so, which is exactly the identification a projection to a new machine cannot use.

The measurement follows that algebra. REML puts λ between 1.4 and 8.1 across the 13 label folds and between 2.0 and 143 across the 11 device folds, which is the spread eleven groups buy, and the held-out score tracks it: 0.264 per label against the power law's 0.214, and **0.829** per device against 0.212, worse on 10 of the 11 devices. Because that optimum is unstable, λ is swept as well as fitted, which bounds what any rule for choosing it could deliver. Over devices the score rises monotonically with λ and is best at $\lambda = 0$, where the estimator is the pooled fit; over labels the best rung is $\lambda = 0.01$ at 0.2127 against 0.2141, a gain of 0.0014. The deployment-matched estimator therefore does not beat a pooled power law on an unseen device even when its one free quantity is chosen against the score it is being judged on. Across the ITER-size-matched cut it does help, at 0.255 against the pooled fit's 0.279, which is the least-squares spelling of the ridge row's 0.278, and it stays well above the constrained fit of Sec. 8 of the main text at 0.183 and the analytic law's 0.194. What has been ruled out is narrower than the class: this is a single random intercept and no random slopes, so a device-level *level* shift is what fails to be the missing structure. Eleven devices is also few enough that the variance component is poorly determined, which is why the sweep rather than the fit carries the conclusion.

Random slopes are the obvious next specification and are worth saying something sharper about than "untested". On the features that carry the size dependence they are not identifiable at all. A device-specific slope is estimated from how the response moves with that feature *within* the device, and Sec. 4.2 of the main text measures how little those features move: major radius carries 0.1% of its log variance within devices, minor radius 1.1%, inverse aspect ratio 4.9% and elongation 6.1%. There is almost no within-device variation for such a slope to be fitted to, which is the same fact that makes those columns close to device labels, read from the other side.

On the four features that do move within a device, plasma current, toroidal field, density and loss power, a random slope is identifiable, and two things stand against fitting one here. Eleven groups that already leave a single variance component spanning 2.0 to 143 will not determine a five-by-five covariance with fifteen free parameters. And the arithmetic that makes the intercept model no better than pooling on an unseen device applies unchanged: a device with no rows has no realised random effect, so it is predicted from the population-level slope whatever the random part is allowed to be, while the random part shifts the population estimate further toward within-device variation, which is the information a new machine cannot use. A richer covariance would have to help through that channel, and the sweep in this section already bounds what that channel can deliver. The specification and the full sweep are in `analysis_mixed_model.py` and `results/mixed_model.json`.

The win counts support an exact test, under a null that is worth naming precisely. Call it the independent-unit-sign null: each unit, label or device as the case may be, is assigned to either model independently with equal probability. Under it, losing 13 of 13 has two-sided $p = 2.4 \times 10^{-4}$ and losing 11 of 11 has $p = 9.8 \times 10^{-4}$. The device-collapsed count is the one to quote, since the 13 database labels are not 13 independent devices, and the same non-independence is a reason to prefer these counts to the 13-label bootstrap.

That null is an idealisation, and Sec. 2 of the main text says in what way: any two of these fits share all but two of their training machines, so the outcomes are correlated through the training set even where the held-out devices are distinct. The descriptive statement carries the claim on its own. The forest is worse on every physical device scored, and the paired interval on the per-device difference excludes zero.

S11 Four choices that do not carry the reversal

The labels too small to score at all. Five labels hold fewer than 30 rows (COMPASS, TCV, START, TDEV and TFTR, 43 rows between them, under 1% of the database) and none can carry a held-out score of its own. They are nonetheless in every training fold, since the threshold governs scoring rather than training, and they are among the most unusual points in the feature space, so whether a flexible model can exploit those few rows is a fair question to ask of the ranking.

Dropping them from training as well answers it. Every fitted model gets slightly worse without them, by 0.012 for the power law, 0.027 for the gradient booster and 0.007 for the random forest, and the ordering over the 13 held-out labels is unchanged: power law 0.214 to 0.226, gradient booster 0.359 to 0.387, random forest 0.465 to 0.472. The 43 rows help the tree ensembles slightly, as one would expect of rows in a sparse region, and nowhere near enough to recover the inversion. The analytic reference is unaffected, since it fits nothing.

Where the row threshold is set. The 13-of-13 count is reported at a 30-row eligibility threshold, which is a choice, so it is swept. Sec. S8 scores leave-one-label-out at 10, 20, 30 and 50 rows. Only which labels are *scored* changes: every label stays in every training fold at every threshold, so the models being compared do not change. The power law wins every scored label at 20, 30 and 50 rows, and 14 of 15 at 10 rows, where the exception is TCV. TCV carries 11 rows, which is why it sits below the threshold in the first place, and a per-label log-RMSE over 11 rows is not a quantity this paper would report on its own. The mean gap moves between +0.211 and +0.251 across the four thresholds and the sign test stays below 10^{-3} throughout, so the headline does not depend on where the line is drawn.

One partition, or any partition. The cross-validation column comes from a single deterministic `GroupKFold` assignment of discharges to five folds. Since this paper is about validation protocol, whether the 29% margin is peculiar to that one partition is a fair question. Repeating the comparison over ten shuffled assignments of the same discharges into the same number of folds gives a mean forest advantage of 27.8% over the power law, ranging from 27.0% to 28.5%, with the forest ahead in all ten. The deterministic partition's 29.5% sits just above that range, so the reported margin is if anything mildly favourable to the forest, and the interpolation win it later gives up is a property of the comparison rather than of one fold assignment.

One ordering inside that sweep is not resolved and the main text says so. The two ensembles are separated by 0.002 in Table 1 of the main text, and across these ten partitions the forest is ahead of the gradient booster in only 2 of 10, and the deterministic partition is one of the two. What every partition agrees on is the power law's position, last of the three under cross-validation and first under leave-one-out, which is the comparison the paper's argument rests on. Which of the two ensembles is nominally best under cross-validation is a coin the partition tosses, and no claim here turns on it.

Tuning the flexible models. Every score above uses the prespecified, untuned settings of Sec. 2 of the main text, which leaves open whether the ensembles lost only because they were not hyperparameter-tuned. That is tested by giving both a grid search nested inside each training fold: three grouped inner folds by discharge over the training rows only, so the held-out discharge fold, label or size cut never takes part in choosing a hyperparameter. The forest searched `max_features` over $\{1.0, 0.5, \mathrm{sqrt}\}$ and `min_samples_leaf` over $\{1, 5\}$, the booster `learning_rate` over $\{0.05, 0.1\}$ and `max_leaf_nodes` over $\{15, 31\}$.

The search does real work on the forest and changes nothing about the conclusion. It moved away from the baseline `max_features` $= 1.0$, `min_samples_leaf` $= 1$ setting in 13 of the 19 folds, and it helps where the paper says a flexible model should help: cross-validation improves from 0.128 to 0.126 and the leave-one-label-out mean from 0.465 to 0.403. At the ITER-size-matched cut it makes the forest *worse*, 0.938 to 1.121, against the power law's 0.278. The booster's search returned scikit-learn's defaults in all 19 folds, so its numbers are unchanged.

The inner folds above group by discharge, which is how a practitioner would ordinarily tune. Grouping them by whole machine instead matches the way the model is deployed, and it helps both models: the forest falls to 0.376 and the booster to 0.350. Both remain well above the power law's 0.214, so the reversal survives either selection procedure.

Tuning therefore widens the interpolation margin that later inverts and does not repair the extrapolation failure, which is what the training-target bound predicts: no setting of these hyperparameters moves a ceiling that comes from averaging training targets.

Two conventions this paper does not follow. Neither the population nor the weighting used above matches how the ITPA scalings were produced, and both are choices rather than findings, so both are measured.

IPB98(y,2) was fitted on a selection that excluded spherical tokamaks; this database does not, and MAST and NSTX are two of the 13 scored labels and two of the four most distant. The main text controls for this only at the ITER-size-matched cut. Dropping every label whose median inverse aspect ratio exceeds 0.5, which removes MAST, NSTX and START and 232 rows, from training and scoring alike, leaves the reversal intact: the forest is worse than the power law on all 11 remaining labels, mean gap +0.182, exact $p = 9.8 \times 10^{-4}$, at 0.395 against the power law's 0.212. The inversion is not an artefact of including the spherical devices.

The weighting arm is the one that qualifies the headline. Every fit above weights rows equally, so JET and ASDEX Upgrade, which supply 77% of them, set most of what each model learns. Refitting every model with each label weighted equally hurts the power law more than the forest, 0.214 to 0.259 against 0.465 to 0.447, and the win count falls from 13 of 13 to **12 of 13** ($p = 3.4 \times 10^{-3}$, mean gap +0.188). The label that changes hands is AUG-W, whose unweighted margin was the second narrowest of the thirteen at +0.012, behind JT-60U's +0.006 which does not change hands, and which moves to −0.067. The direction and its significance survive the reweighting; unanimity does not, and the 13 of 13 quoted throughout is the unweighted count.

S12 The same distance in dimensionless coordinates

Organising anything in this field by engineering variables invites one objection, and it is the objection Hall *et al.* [17] have raised in a conference presentation: the weak size dependence in this database may come from a subset localised in *dimensionless* space rather than from size, in which case a distance in engineering space measures a shadow of the thing that matters. That is checkable here rather than only citable. The Connor–Taylor constraints of Sec. 8 of the main text are derived in code from the scalings of ρ^*, β and ν^*, so the group definitions are already written down, and the stored energy joined across in Sec. 4.2 of the main text supplies the temperature they need. Placing every row by

$$\rho^* \sim T^{1/2}B^{-1}L^{-1}, \qquad \beta \sim nTB^{-2}, \qquad \nu^* \sim nLT^{-2}, \tag{3}$$

with $T \sim W_{\mathrm{th}}/(nV)$ and $V \sim Ra^2\kappa$, and repeating the distance construction above on those three coordinates instead of the nine, gives a second ranking of the 13 labels. Multiplicative constants are dropped throughout and cannot matter, since every quantity here is a difference of mean log-vectors.

The two rankings agree at $\rho = +0.78$, and the diagnosis does not depend on which is used. On the 6214 rows a dimensionless placement is possible for, the forest's per-label error tracks dimensionless distance at $\rho = +0.77$ against $+0.79$ for engineering distance, the booster at $+0.35$ against $+0.51$, and the power law is flat against both, -0.04 and -0.06. The headline $+0.85$ is the engineering figure on all 6228 rows; dropping the 14 rows with no stored energy moves it to $+0.79$, and the coordinate change then moves it by 0.02. The distance reading above survives translation into the coordinates the objection is stated in.

The size cut does not fare as well, and this is where the objection lands. At the ITER-size-matched cut the two sides are separated in dimensionless space as well as in size: the interquartile ranges of $\log \rho^*$ do not overlap at all and its median moves by 1.21 training standard deviations, $\log \nu^*$ overlaps by 0.26 at 0.85 standard deviations, and only $\log \beta$ stays put, overlapping by 0.69 at 0.22. Larger machines in this database run at smaller normalised gyroradius and lower collisionality, which is physics rather than an artefact, and it means the cut is not a size boundary with everything else held fixed. Every result scored across it is therefore transfer across a joint displacement in size, ρ^* and ν^*, and the paper says so rather than describing it as a size extrapolation alone. The leave-one-device-out comparison of Sec. 4 of the main text carries no such confound, which is the second reason it and not the cut is the claim this paper rests on.

S13 What the elongation substitution costs

ITPA20 and ITPA20-IL are specified on the separatrix elongation at the geometric axis, κ, and STD5 delivers one elongation column, `KAPPAA`: the areal elongation $\kappa_a = S/(\pi a^2)$, which is the definition IPB98(y,2) was fitted to. Sec. 4.2 of the main text scores each law on the column it was fitted to, taking κ from the full DB5.2.3 revision. What that substitution is worth is measured here rather than modelled, and the measurement is also the argument for not modelling it.

On a parametrised boundary κ/κ_a depends on the triangularity alone, which confines the ratio to $[1.000, 1.110]$. The measured ratio has a median of **1.083** and a maximum of **1.341**, with 16% of rows above that ceiling and 7% *below* unity, which no boundary of that family produces at all. Only **2.0%** are below it by more than 5×10^{-4}; the rest is rounding on machines that ran circular.

The rows below unity are worth following, because the obvious reading of them is that the delivered column is wrong and the obvious reading is not right here. A boundary elongation below the areal elongation is impossible for a *convex* boundary, and of the 2.0% that clear rounding, all but PBX-M's 59 rows sit within 3.5% of unity on near-circular machines, which the two columns' precision covers. PBX-M does not: its ratio runs from 0.67 to 0.80, at a median of **0.739**. PBX-M ran indented, bean-shaped plasmas, its boundary is not convex, and the convexity argument does not apply to it. DB5.2.3 records that directly: of the analysed rows, PBX-M is the only label carrying an indentation above 0.04, at a median of **0.154**, and the 60 rows with any indentation at all have a median κ/κ_a of 0.740. The column is not defective on those rows; the parametrised boundary is simply the wrong shape for them, which is the same point the ratio makes everywhere else and is here checkable against a third column.

Scoring on the delivered column rather than the measured one moves ITPA20 by up to 0.018 and ITPA20-IL by up to 0.022, twenty to thirty times what the shape model would predict. Where a column exists, a model of it is not evidence about it.

References

[1] P.N. Yushmanov *et al.*, "Scalings for tokamak energy confinement," *Nucl. Fusion* **30**, 1999 (1990). ITER89-P. doi:10.1088/0029-5515/30/10/001

[2] ITER Physics Expert Group on Confinement and Transport, ITER Physics Expert Group on Confinement Modelling and Database and ITER Physics Basis Editors, "Chapter 2: Plasma confinement and transport," *Nucl. Fusion* **39**, 2175 (1999). doi:10.1088/0029-5515/39/12/302

[3] E. J. Doyle *et al.*, "Chapter 2: Plasma confinement and transport," *Progress in the ITER Physics Basis*, *Nucl. Fusion* **47**, S18 (2007). doi:10.1088/0029-5515/47/6/S02

[4] A. Loarte *et al.*, "The new ITER baseline, research plan and open R&D issues," *Plasma Phys. Control. Fusion* **67**, 065023 (2025). doi:10.1088/1361-6587/add9c9

[5] A.J. Creely *et al.*, "Overview of the SPARC tokamak," *J. Plasma Phys.* **86**, 865860502 (2020). doi:10.1017/S0022377820001257

[6] JT-60SA Research Unit, *JT-60SA Research Plan: Research Objectives and Strategy*, Version 5.0 (2021). `https://www.jt60sa.org/wp/wp-content/uploads/2021/02/JT-60SA_Res_Plan-5.pdf`

[7] M. Yoshida *et al.*, "Overview of plasma operation and control studies in the first plasma campaign of JT-60SA," *Plasma Phys. Control. Fusion* **67**, 065010 (2025). doi:10.1088/1361-6587/add563

[8] J. Garcia *et al.*, "Overview of first JT-60SA plasma operation and plans in view of ITER and DEMO," *Nucl. Fusion* **66**, 116011 (2026). doi:10.1088/1741-4326/ae74e1

[9] M. Kleiber, "Body size and metabolism," *Hilgardia* **6**, 315 (1932). doi:10.3733/hilg.v06n11p315

[10] J. Felsenstein, "Phylogenies and the comparative method," *Am. Nat.* **125**, 1 (1985). doi:10.1086/284325

[11] D.S. Falster *et al.*, "BAAD: a Biomass And Allometry Database for woody plants," *Ecology* **96**, 1445 (2015). Released CC0. doi:10.1890/14-1889.1

[12] G.B. West, J.H. Brown and B.J. Enquist, "A general model for the structure and allometry of plant vascular systems," *Nature* **400**, 664 (1999). doi:10.1038/23251

[13] G. Verdoolaege *et al.*, "The updated ITPA global H-mode confinement database: description and analysis," *Nucl. Fusion* **61**, 076006 (2021). doi:10.1088/1741-4326/abdb91

[14] L. Breiman, "Random forests," *Machine Learning* **45**, 5 (2001). doi:10.1023/A:1010933404324

[15] Z. Wang *et al.*, "Power-law-anchored residual learning for H-mode energy confinement time in tokamaks: interpolation and parameter-defined extrapolation," arXiv:2608.20684 (2026). doi:10.48550/arXiv.2608.20684

[16] D.R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography* **40**, 913 (2017). doi:10.1111/ecog.02881

[17] J. Hall, P. Zhang, Y. Zhang, C. Angioni and G. Verdoolaege, "Confinement scaling with machine size in the updated ITPA global H-mode energy confinement database," poster P2.1097, *52nd EPS Conf. on Plasma Physics*, Edinburgh (2026). Conference presentation; not published in refereed form at the time of writing.